

2ai	The <u>acceleration of the block is directly proportional to its displacement from its equilibrium position, and the acceleration is always directed towards that position as indicated by the negative sign.</u>	B1 B1
aii	From graph, $T = 4 \text{ s}$ $f = \frac{1}{T} = \frac{1}{4} = 0.25 \text{ Hz}$	C1 A1
Aiii	Comparing $a = -\omega^2 x$ and $a = -\frac{28}{m} x$, $\omega^2 = \frac{28}{m}$ $4\pi^2 f^2 = \frac{28}{m}$ $m = 11.3 \text{ kg}$	C1 A1
aiv	Max. k.e. = $\frac{1}{2} m \omega^2 x_o^2 = 1.8$ $x_o = 0.36 \text{ m}$	C1 A1
b	displacement / m 0.36 0 -0.36 4 8 time / s	B1 Cosine -cosine B1 labels